

Xinyan Wang

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RESEARCH INTERESTS

LLM post-training (RL, on-policy distillation), LLM agents and agent red-teaming, efficient inference.

EDUCATION

- **University of Wisconsin–Madison** Madison, WI, USA
Ph.D. in Statistics; Advisor: Prof. Jun Shao Sep 2023 – May 2028 (Expected)
M.S. in Computer Science Sep 2023 – May 2026
M.S. in Statistics Sep 2022 – May 2023
Visiting Student in Statistics and Computer Science Sep 2021 – May 2022
- **East China Normal University** Shanghai, China
B.S. in Statistics Sep 2018 – June 2022

EXPERIENCE

- **SaFo Lab** Johns Hopkins University
Research Intern; Mentor: Prof. Chaowei Xiao Aug 2025 – Present
 - Research on safe and efficient large reasoning models and LLM agents.

RESEARCH EXPERIENCE

- **ROM: Real-time Overthinking Mitigation in Large Reasoning Models** JHU
First author; advised by Prof. Chaowei Xiao (Submitted to ARR, 2026)
 - Showed overthinking in large reasoning models is a latent, decodable productive-to-redundant transition, and built ROM, a streaming detector ($\sim 0.1\%$ of parameters) intervening on a frozen LRM in real time.
 - Attained the highest accuracy in 19 of 25 settings across five backbones and five benchmarks, cutting response length 28–77% and latency 46.5%.
- **ReasoningBomb: A Denial-of-Service Attack on Large Reasoning Models** JHU
With Prof. Chaowei Xiao et al. (ACM CCS 2026)
 - Formalized prompt-induced inference-time denial-of-service (PI-DoS) for LRMs and built a reinforcement-learning attacker that crafts short, natural prompts inducing pathologically long reasoning.
 - Across 7 open-source and 3 commercial models, induced $286.7\times$ input-to-output amplification and bypassed input-, output-, and dual-stage detectors at 99.8%, 98.7%, and 98.4%.
- **PW-OPSD: Position-Weighted On-Policy Self-Distillation for Reasoning** JHU
With Prof. Chaowei Xiao et al. (Submitted to NeurIPS 2026)
 - Built a branch-viability diagnostic that forces privileged-teacher token alternatives after the student’s on-policy prefix and tests whether the correct answer still emerges.
 - Showed on Qwen3-4B that within-sequence position predicts teacher-token reliability far better than local uncertainty (AUROC 0.83 vs. 0.57); the resulting PW-OPSD up-weights later tokens, improving AIME 2024/2025 Avg@12 by +1.0/+1.1 at no extra teacher compute.
- **MLE with Datasets from Populations Having Shared Parameters** UW–Madison
Advisor: Prof. Jun Shao (Statistical Theory and Related Fields, 2023)
 - Studied maximum likelihood estimation with multiple datasets from heterogeneous populations sharing parameters; established asymptotic normality, estimated variability via bootstrap, and reduced the estimator’s standard deviation by nearly 57% versus single-dataset MLEs.

PUBLICATIONS

Xiaogeng Liu, **Xinyan Wang**, Yechao Zhang, Sanjay Kariyappa, Chong Xiang, Muhao Chen, G. Edward Suh, Chaowei Xiao. *ReasoningBomb: A Stealthy Denial-of-Service Attack by Inducing Pathologically Long Reasoning in Large Reasoning Models*. ACM Conference on Computer and Communications Security (CCS), 2026.

Xinyan Wang, Xiaogeng Liu, Ming Pei, Chaowei Xiao. *ROM: Real-time Overthinking Mitigation via Streaming Detection and Intervention*. Submitted to ACL Rolling Review (ARR), 2026. arXiv:2603.22016.

Xiaogeng Liu, **Xinyan Wang**, Yingzi Ma, Yechao Zhang, Chaowei Xiao. *When Are Teacher Tokens Reliable? Position-Weighted On-Policy Self-Distillation for Reasoning*. Submitted to NeurIPS 2026. arXiv:2605.21606.

Jun Shao, **Xinyan Wang**. *MLE with Datasets from Populations Having Shared Parameters*. Statistical Theory and Related Fields, 7(3):213–222, 2023.

TEACHING

STAT 240 (Data Science Modeling I), STAT 309 (Introduction to Probability and Mathematical Statistics I), STAT 340 (Data Science Modeling II), STAT 610 (Statistical Inference), STAT 611 (Statistical Models for Data Science), STAT 613 (Statistical Methods for Data Science).

PROFESSIONAL SERVICE

Reviewer: ACL 2026, ECCV 2026, EMNLP 2026, NeurIPS 2026

SKILLS

Languages: Python, R, SQL

Frameworks: PyTorch, Scikit-learn, Flask